Tiffany Erin Ta, BHSc

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EDUCATION

PhD – Department of Biochemistry and Biomedical Sciences
2024 – 2028 (expected graduation year)
McMaster University, Hamilton, ON

- Research interest: bioinformatics – artificial intelligence, natural language processing.
- PhD Thesis (McArthur Lab) – leveraging natural language processing for antimicrobial resistance surveillance.
- Courses completed:
  - Biochemistry 732 – Writing for Science
  - Biochemistry 733 – Biomedical Commercialization and Entrepreneurship

Honours Bachelor of Health Sciences – Biomedical Discovery and Commercialization
2020 - 2024
McMaster University, Hamilton, ON

- Completed an undergraduate thesis in the McArthur Lab (see **Projects** for more information).
- Courses
  - Organic chemistry, drug discovery and development, research skills laboratory and inquiry, accounting, organizational behaviour, finance, marketing, proteins and nucleic acids, analytical chemistry, and physical chemistry.

AWARDS AND SCHOLARSHIPS

1. **Faculty of Health Sciences Graduate Programs Outstanding Achievement Award.** McMaster University, April 2026.

2. **Best Poster Award: Antimicrobial Resistance – Genomes, Big Data, and Emerging Technologies.** Wellcome Genome Campus, Hinxton, Cambridgeshire, England. March 2026.

3. **Scholarship ($12,000): Nominated for the Faculty Ontario Graduate Fellowship.** McMaster University, Sept 2024.

4. **Scholarship ($4,000): David Braley Center for Antibiotic Discovery (DBCAD) Summer Fellowship.** McMaster University, May 2024.

5. **Scholarship ($4,000): David Braley Center for Antibiotic Discovery (DBCAD) Summer Fellowship.** McMaster University, May 2023.

6. **Scholarship ($3,000): McMaster Entrance Scholarship.** McMaster University, Sept 2020.

INVITED TALKS

1. **Tiffany E. Ta**, Andrew G. McArthur. *Using Large Language Models for Antimicrobial Resistance Surveillance.* Zoonoses, Antimicrobial Resistance, and Bioinformatics (ZAMBI). Dalhousie University, Halifax, Nova Scotia, Canada. July 2026.

2. **Tiffany E. Ta**, Andrew G. McArthur. *Applying Natural Language Processing for Antimicrobial Resistance Surveillance.* BIOCHEM 3BP3 (Undergraduate Bioinformatics Course) Guest Lecturer. McMaster University, Hamilton, Ontario, Canada. October 2025.

3. **Tiffany E. Ta**, Andrew G. McArthur. *Named entity recognition and relationship extraction to mine minimum inhibitory concentration of antibiotics from biomedical text.* Intelligent Systems for Molecular Biology and the European Conference on Computational Biology. ACC Liverpool, Liverpool, England. July 2025.

4. **Tiffany E. Ta**, Andrew G. McArthur. *Using Deep Learning for Text Extraction of Minimum Inhibitory Concentration of Antibiotics.* Biochemistry and Biomedical Sciences Research Symposium. McMaster University, Hamilton, Ontario, Canada. March 2025.

5. **Tiffany E. Ta**, Andrew G. McArthur. *Using Natural Language Processing to Analyze and Visualize the Transmission Pathways of Antimicrobial Resistance Genes.* Biomedical Discovery and Commercialization Engage. McMaster University, Hamilton, Ontario, Canada. March 2024.

POSTER PRESENTATIONS

1. **Tiffany E. Ta**, Brian P. Alcock, Andrew G. McArthur. *Using Natural Language Processing for Antimicrobial Resistance Surveillance.* Antimicrobial Resistance – Genomes, Big Data and Emerging Technologies. Wellcome Genome Campus, Hinxton, Cambridgeshire, England. March 2026.

2. **Tiffany E. Ta**, Brian P. Alcock, Andrew G. McArthur. *CARD*CURATE: Developing Tools for an Artificial Intelligence Agent to Expedite the Curation of Antibiotic and Antimicrobial Resistance Gene Knowledge*. Institute of Infectious Disease Research Trainee Day. McMaster University, Hamilton, Ontario, Canada. November 2025.

3. Trisha Parchoor, **Tiffany E. Ta**, Brian P. Alcock, Andrew G. McArthur. *Using Genomics and Natural Language Processing to Assess Antibiotic Resistant Gene Risk*. Women in Science and Engineering Conference. McMaster University, Hamilton, Ontario, Canada. March 2025.

4. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing for Antimicrobial Resistance Epidemiology and Surveillance.* Emerging Pandemics and Infections Consortium AMR Symposium. Sheraton Center Hotel, Toronto, Ontario, Canada. November 2024.

5. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing to Extract Minimum Inhibitory Concentration of Antibiotics from PubMed Central.* Institute of Infectious Disease Research Trainee Day. McMaster University, Hamilton, Ontario, Canada. November 2024.

6. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing to Analyze and Visualize the Transmission Pathways of Antimicrobial Resistance Genes.* Canadian Society of Microbiologists. Western University, London, Ontario, Canada. June 2024.

7. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing to Analyze and Visualize the Transmission Pathways of Antimicrobial Resistance Genes.* Antimicrobial Resistance – Genomes, Big Data and Emerging Technologies. Wellcome Genome Campus, Hinxton, Cambridgeshire, England. March 2024.

8. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing to Analyze and Visualize the Transmission Pathways of Antimicrobial Resistance Genes.* Biochemistry and Biomedical Sciences Research Symposium. McMaster University, Hamilton, Ontario, Canada. March 2024.

9. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing to Analyze the Transmission Pathways of Antimicrobial Resistance Genes Through a "Confusogram".* Canadian Undergraduate Conference on Healthcare. Queens University, Kingston, Ontario, Canada. January 2024.

10. **Tiffany E. Ta**, Arman Edalatmand, Andrew G. McArthur. *Using Natural Language Processing to Analyze the Transmission Pathways of Antimicrobial Resistance Genes Through a "Confusogram".* Institute of Infectious Disease Research Trainee Day. McMaster University, Hamilton, Ontario, Canada. November 2023.

PUBLICATIONS

1. Wlodarski, M. A. *et al.* CARD k-mers: Unmasking the pathogen hosts and genomic contexts of antimicrobial resistance genes in metagenomic sequences. *bioRxiv* 2025.09.15.676352 (2025) doi:10.1101/2025.09.15.676352. [SUBMITTED]
2. Edalatmand A., Ta, T. E, *et al*. CARD:Epi – Contextualizing Antimicrobial Resistance Determinants Using Deep Learning Language Models. [MANUSCRIPT IN PREPARATION]
3. Ta, T. E, *et al.* CARD:MIC – Leveraging Natural Language Processing to Extract Minimum Inhibitory Concentration of Antibiotics from Biomedical Text. [MANUSCRIPT IN PREPARATION]

SKILLS/CERTIFICATIONS

- Capable of working with Microsoft Word, Excel, PowerPoint, and Teams.
- Completed Introduction to Python (DataCamp).
- Completed Intermediate Python (DataCamp).

EXPERIENCE

Teaching Assistant (McArthur Lab) Sept 2025 – Dec 2025
McMaster University, Hamilton

- Teaching assistant for BIOCHEM 3BP3 (Practical Bioinformatics in the Genomics Era).
- Deliver tutorials on a weekly-basis, providing in-depth demonstrations and walkthroughs for their laboratory assignments.

Developer Assistant (McArthur Lab) May 2024 – Aug 2024
McMaster University, Hamilton

- Bioinformatics lab in the Michael G. DeGroote Institute for Infectious Disease Research.
- Used large language models for full-text analysis of biomedical publications.

Developer Assistant (McArthur Lab) May 2023 – Aug 2023
McMaster University, Hamilton

- Bioinformatics lab in the Michael G. DeGroote Institute for Infectious Disease Research.
- Used natural language processing techniques to visualize big data, creating a novel antimicrobial resistance "Confusogram".

PROJECTS

Using Natural Language Processing for Antimicrobial Resistance Surveillance (PhD Project)
Sept 2024 – Present

- Using ~6 million papers from PubMed Central, extract minimum inhibitory concentration (MIC) values of antibiotics using large language models.
- Use natural language processing (NLP) to extract meaningful biomedical text from papers.
- Built a machine learning classifier to identify bacterial drug resistance papers.
- Perform named entity recognition and relation extraction to annotate qualitative MIC data.

- Use a combination of Regular Expressions and generative models to extract quantitative MIC data in sentences and tables.
- Develop a multi-agentic framework for AI-assisted biocuration.
- Supervisor: Dr. Andrew G. McArthur
- Committee Members: Dr. Jonathan M. Stokes, Dr. Gerard D. Wright

Developing an Artificial Intelligence Agent for AllTheBacteria (ESGEM-AMR Hackathon at Jesus College, Cambridge University, England) Mar 2026

- https://github.com/AMR-genomics-hackathon-2026/atb-ai-agent
- AllTheBacteria is an open-access resource containing bacterial and archaeal genomes.
- I developed an AI agent that has AllTheBacteria knowledge, queries PubMed, and performs sequence analysis (e.g., identifies multi-locus sequence types).

Comparing Transformers for Biomedical Text Extraction from PubMed Central Articles (DBCAD Summer Fellowship) May 2024 – Aug 2024

- Train models (BioBERT, BioM-ALBERT, BiomedBERT) for named entity recognition to extract biomedical text and compare evaluation metrics to determine the best-performing model.
- Supervisor: Dr. Andrew G. McArthur

Using Natural Language Processing to Analyze the Transmission of Antimicrobial Resistance Genes (DBCAD Summer Fellowship and Undergraduate Thesis Project) May 2023 – May 2024

- https://github.com/tiffanyta1402/cardepi-analysis
- Previous MSc work in the McArthur Lab involved generating a dataset using machine learning regarding epidemiology and antibiotic resistance gene terms.
- I have generated 2 pipelines that produce visuals using the python library Graphviz (to create a confusogram network) and Uniform Manifold Approximation and Projection (UMAP) to help understand different pathways in which antimicrobial resistance genes can be transmitted.
- My work incorporates an interdisciplinary analysis of data across scientific (NLP, genetics of AMR), sociopolitical (adjusting for different national approaches to antibiotic use or food production), and environmental (climate, wastewater fate) perspectives.
- Future work involves using NLP tasks (i.e., named entity recognition and relationship extraction) to extract meaningful biomedical text from published papers from PubMed.
- Supervisor: Dr. Andrew G. McArthur

MACtion Potential Projects Sept 2020 – April 2024

- 2023-24: Created a mood-controlled lamp. Based on the user's emotions (happy or angry), the lamp will turn a specific colour based on EEG signals. Led various subteams (i.e., software, mechanical, research, electrical, etc…) to achieve their goals within this project. Guide research subteam for this project.
- 2022-23: Analyzed withdrawal symptom EEG signals. Conducted research on how drugs and withdrawal affect the brain while leading various subteams (i.e., software, mechanical, research, electrical, etc…) to achieve their goals within this project.
- 2021-22: Collaborated with other members to create a blink-controlled car project using an Arduino and machine learning (a support vector machine model).
- 2020-21: Used EEG analysis from a Muse headband to monitor students' attention while watching a "confusing" video. Helped to research ways to read, reduce, and prevent artifacts from arising in the Muse headband.

MENTORSHIP EFFORTS

McArthur Lab Undergraduate Mentorship

1. **Sophia Green** (Undergraduate Thesis Student) Sept 2025 – Apr 2026

An Investigation into Workflow Construction and Generalization Performance for an Integrated Approach to Biomedical Text Mining

2. **Nilasha Mohan** (Undergraduate Thesis Student) May 2024 – Apr 2025
Developing Hidden Markov Models for the Detection and Annotation of Beta-Lactamases in Metagenomic Sequences.

3. **Daljot Sidhu, Sarina Badre Rajaei, Melody Ding, Lucas Fong, Yufei Liu, Zahrah Talawala, Jiaying Liu** (Undergraduate McArthur Lab volunteers) Oct 2024 – Apr 2025
Identification of Bacterial Drug Resistance Papers to Compare to a Machine Learning Model.

MACtion Potential Mentorship Sept 2022 – May 2024

- Advised undergraduate students in the club on project decisions.
- Provide feedback to students on their project ideas and shared knowledge on different topics (e.g., bridging the gaps in engineering concepts to neuroscience).
- Challenged students to solve problems using innovative ideas.

EXTRACURRICULAR ACTIVITIES

Co-President of MACtion Potential Sept 2022 – May 2024

- NeurotechX affiliated club at McMaster University. The club helps to enhance students' understanding of neuroscience, neurotechnology, computer science, and research skills.
- Designed a website for the club: https://sites.google.com/view/maction-potential/home
- Managed the projects for the 2022-23 and 2023-24 school years. Provide guidance and instructions to different subteams within the club (software, mechanical, signal processing, research, and electrical) to create a neurotechnology device.
- Led administrative tasks and delegated tasks to executive members.

Team Meta-Model Volunteer (McArthur Lab) Jan 2023 – Apr 2023

- Volunteering in Dr. Andrew McArthur's lab at the Michael DeGroote Centre for Learning and Discovery, aiding in curation of the Comprehensive Antibiotic Resistance Database and researching vancomycin cluster meta-models.
- Aided in the curation of antibiotic resistant genes.

BioTEC Ambassador July 2021 – July 2023

- Helped with promoting the annual University of Waterloo BioTEC conference.
- Worked with a small team to create BioTEC's first sticker contest.
- Created social media posts to bring awareness to the conference.